

Exercise Sheet 1

Exercise 1. Prove the lower bound $\pi(x) \geq \log \log x$ for the number of primes up to x . Hint: if p_n is the n -th prime, show $p_n \leq 2^{2^{n-1}}$, hence $p_n \leq e^{e^{n-1}}$. *Solution.* Let p_n be the n -th prime. We prove by induction that $p_n \leq 2^{2^{n-1}}$. The case $n = 1$ is clear. If the claim holds up to n , Euclid's argument gives $p_{n+1} \leq p_1 \cdots p_n + 1 \leq 2^{1+2+\cdots+2^{n-1}} + 1 = 2^{2^n - 1} + 1 \leq 2^{2^n}$. Hence $p_n \leq 2^{2^{n-1}} \leq e^{e^{n-1}}$. If $x \geq e$, take $n = \lfloor \log \log x \rfloor + 1$; then $p_n \leq x$, so $\pi(x) \geq n \geq \log \log x$. For $x < e$ the right side is ≤ 0 or the assertion is immediate.

Exercise 2. Prove: (a) $\tau(n^2) = \sum_{d|n} \mu(d) \tau(n/d)^2$; (b) $\sum_{d^2|n} \mu(d) = \mu(n)^2$ and $\sum_{d^k|n} \mu(d) = 0$ if $m^k \mid n$ for some $m > 1$, and 1 otherwise; (c) $\mu(n) = \sum_{1 \leq a \leq n, (a,n)=1} e^{2\pi i a/n}$. *Solution.* (a) Both sides are multiplicative in n . At $n = p^a$, the right side is $\tau(p^a)^2 - \tau(p^{a-1})^2 = (a+1)^2 - a^2 = 2a+1 = \tau(p^{2a})$. (b) Since $\mu(d)$ vanishes unless d is square-free, $\sum_{d^2|n} \mu(d) = \prod_{p^2|n} (1-1)$, with the empty product 1; this is $\mu(n)^2$. Similarly $\sum_{d^k|n} \mu(d) = \prod_{p^k|n} (1-1)$, hence it is 0 exactly when some $m^k \mid n$ with $m > 1$, and otherwise 1. (c) Using $1_{(a,n)=1} = \sum_{d|(a,n)} \mu(d)$, we get $\sum_{1 \leq a \leq n, (a,n)=1} e^{2\pi i a/n} = \sum_{d|n} \mu(d) \sum_{1 \leq b \leq n/d} e^{2\pi i b/(n/d)}$. The inner geometric sum is 0 unless $d = n$, and then it is 1, so the result is $\mu(n)$.

Exercise 3. Prove $\sum_{d|n} \mu(d)^2 = 2^{\omega(n)}$ and $\sum_{d|n} \mu(d) \tau(d) = (-1)^{\omega(n)}$. *Solution.* The first sum counts square-free divisors of n , hence one may choose independently whether each of the $\omega(n)$ prime divisors occurs, giving $2^{\omega(n)}$. The second sum is multiplicative in n ; at p^a it equals $1 - \tau(p) = 1 - 2 = -1$, so multiplying over all distinct prime divisors gives $(-1)^{\omega(n)}$.

Exercise 4. Let f be multiplicative. Show that f^{-1} is completely multiplicative iff $f(p^\ell) = 0$ for all $\ell \geq 2$. *Solution.* Let $g = f^{-1}$ and use local generating series $F_p(T) = \sum_{\ell \geq 0} f(p^\ell) T^\ell$ and $G_p(T) = \sum_{\ell \geq 0} g(p^\ell) T^\ell$, with $F_p G_p = 1$. If g is completely multiplicative, then $G_p(T) = \sum_{\ell \geq 0} g(p)^\ell T^\ell = (1 - g(p)T)^{-1}$, so $F_p(T) = 1 - g(p)T$ and $f(p^\ell) = 0$ for $\ell \geq 2$. Conversely, if $F_p(T) = 1 + f(p)T$ for every p , then $G_p(T) = (1 + f(p)T)^{-1} = \sum_{\ell \geq 0} (-f(p))^\ell T^\ell$, so $g(p^\ell) = (-f(p))^\ell$; together with multiplicativity of g , this is complete multiplicativity.

Exercise 5. Show that there are exactly two arithmetic functions f with $f * f = u$. Are they multiplicative? *Solution.* From $n = 1$ we get $f(1)^2 = 1$, so $f(1) = \pm 1$. For each $n > 1$, the equation is $2f(1)f(n) + \sum_{d|n, 1 < d < n} f(d)f(n/d) = 1$, which determines $f(n)$ recursively from earlier values. Thus there are exactly two solutions. The solution with $f(1) = 1$ has Dirichlet series $\zeta(s)^{1/2} = \prod_p (1 - p^{-s})^{-1/2}$, hence is multiplicative, with local coefficients $\binom{2a}{a}/4^a$ at p^a . The other solution is its negative; since its value at 1 is -1 , it is not multiplicative in the usual sense.

Exercise 6. Assume $f(1) \neq 0$ and $g(1) \neq 0$. Prove $(f * g)^{-1} = f^{-1} * g^{-1}$. *Solution.* By commutativity and associativity of Dirichlet convolution, $(f * g) * (f^{-1} * g^{-1}) = (f * f^{-1}) * (g * g^{-1}) = \varepsilon * \varepsilon = \varepsilon$, where $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$. Hence $f^{-1} * g^{-1}$ is the Dirichlet inverse of $f * g$.

Exercise 7. For $k \geq 0$, define $\Lambda_k(n) = \sum_{d|n} \mu(d) (\log(n/d))^k$. Prove $\Lambda_k(n) = 0$ if $\omega(n) > k$, and $0 \leq \Lambda_k(n) \leq (\log n)^k$. *Solution.* Let $p_1, \dots, p_r$ be the distinct prime divisors of n , put $X = \log n$ and $y_i = \log p_i$. Then $\Lambda_k(n) = \sum_{S \subset \{1, \dots, r\}} (-1)^{|S|} (X - \sum_{i \in S} y_i)^k$, the r -fold finite difference of the polynomial X^k . If $r > k$, this is 0. If $r \leq k$, repeated integration gives $\Lambda_k(n) = k(k-1) \cdots (k-r+1) \int_0^{y_1} \cdots \int_0^{y_r} (X - t_1 - \cdots - t_r)^{k-r} dt_r \cdots dt_1 \geq 0$. Finally $1 * \Lambda_k = (1 * \mu) * (\log^k) = \log^k$, so $(\log n)^k = \sum_{d|n} \Lambda_k(d) \geq \Lambda_k(n)$.

Exercise Sheet 2

Exercise 1. Find an arithmetic function f such that, for all $n \geq 1$, $1/\varphi(n) = \sum_{d|n} d^{-1} f(n/d)$. *Solution.* Let $a(n) = 1/n$. Then $1/\varphi = a * f$, so $f = a^{-1} * (1/\varphi)$ and $a^{-1}(n) = \mu(n)/n$. Thus f is multiplicative. For p^α , $f(p) = 1/(p-1) - 1/p = 1/(p(p-1))$ and $f(p^\alpha) = 1/(p^{\alpha-1}(p-1)) - 1/(p^{\alpha-1}(p-1)) = 0$ for $\alpha \geq 2$. Hence $f(1) = 1$ and, for $n > 1$, $f(n) = \mu(n)^2 \prod_{p|n} 1/(p(p-1))$.

Exercise 2. Show that there are constants c_1, c_2 such that, for all $x \geq 1$, $\sum_{n \leq x} 1/\varphi(n) = c_1 \log x + c_2 + O(\log x/x)$. *Solution.* Using Exercise 1, $\sum_{n \leq x} 1/\varphi(n) = \sum_{m \leq x} f(m) H_{\lfloor x/m \rfloor}$, where $H_y = \log y + \gamma + O(1/y)$. The Euler product for f gives $\sum_{m > x} f(m)(1 + \log m) \ll \log x/x$ and $\sum_{m \leq x} mf(m) \ll \log x$. Therefore $\sum_{n \leq x} 1/\varphi(n) = \sum_{m \leq x} f(m) \{\log x - \log m + \gamma\} + O(\log x/x) = c_1 \log x + c_2 + O(\log x/x)$, where $c_1 = \sum_{m \geq 1} f(m)$ and $c_2 = \gamma c_1 - \sum_{m \geq 1} f(m) \log m$.

Exercise 3. Show that $\sum_{2 < n \leq x} 1/(n \log n) = \log \log x + C_1 + O(1/(x \log x))$ for $x \geq 3$. *Solution.* Apply Euler–Maclaurin, or the integral estimate for a monotone C^1 function, to $h(t) = 1/(t \log t)$ on $[3, \infty)$. Since $\int_3^x h(t) dt = \log \log x - \log \log 3$ and the tail of the correction term is $O(h(x)) = O(1/(x \log x))$, the difference between the sum and the integral tends to a constant $C_1 + \log \log 3$, giving the stated formula.

Exercise 4. For $r_2(n) = \#\{(a, b) \in \mathbb{Z}^2 : a^2 + b^2 = n\}$, show $\sum_{n \leq x} r_2(n) = \pi x + O(\sqrt{x})$. *Solution.* The sum counts lattice points in the closed disc $D_R = \{(u, v) : u^2 + v^2 \leq R^2\}$ with $R = \sqrt{x}$. Attach to each lattice point the unit square centered at it. The union differs from D_R only inside a strip of bounded width around the circle of radius R , whose area is $O(R)$. Hence the number of lattice points is $\text{area}(D_R) + O(R) = \pi R^2 + O(R) = \pi x + O(\sqrt{x})$.

Exercise 5. An integer is square-free if it is not divisible by any square. Let $Q_2(x) = \#\{n \leq x : n \text{ is square-free}\}$. Show $Q_2(x) = C_2 x + O(\sqrt{x})$ for some constant C_2 . *Solution.* Since $\mu(n)^2 = \sum_{d^2|n} \mu(d)$, we have $Q_2(x) = \sum_{d \leq \sqrt{x}} \mu(d) \lfloor x/d^2 \rfloor = x \sum_{d \leq \sqrt{x}} \mu(d)/d^2 + O(\sqrt{x})$. Extending the absolutely convergent sum gives an additional error $x \sum_{d > \sqrt{x}} d^{-2} = O(\sqrt{x})$. Thus $C_2 = \sum_{d \geq 1} \mu(d)/d^2 = 1/\zeta(2)$.

Exercise 6. For every $\varepsilon > 0$, show that there is $C_\varepsilon > 0$ such that $\tau(n) \leq C_\varepsilon n^\varepsilon$ for all $n \geq 1$, where τ is the divisor function. *Solution.* Choose P so large that $a+1 \leq p^{a\varepsilon}$ for all $a \geq 1$ and all primes $p > P$. For the finitely many primes $p \leq P$, set $C_\varepsilon = \prod_{p \leq P} \sup_{a \geq 0} (a+1)p^{-a\varepsilon} < \infty$. If $n = \prod p^{a_p}$, then $\tau(n) = \prod (a_p + 1) \leq C_\varepsilon \prod p^{a_p \varepsilon} = C_\varepsilon n^\varepsilon$.

Exercise Sheet 3

Exercise 1. Let f satisfy $\sum_{p \leq x} f(p) \log p = (ax + b) \log x + cx + O(1)$ for $x \geq 2$. Prove that for some constant B , $\sum_{p \leq x} f(p) = ax + (a+c)(x/\log x + \int_2^x dt/\log^2 t) + b \log \log x + B + O(1/\log x)$. *Solution.* Let $A(x) = \sum_{p \leq x} f(p) \log p$. Abel summation gives $\sum_{p \leq x} f(p) = A(x)/\log x + \int_2^x A(t)/(t \log^2 t) dt$. Insert the hypothesis. The $O(1)$ term contributes a constant plus $O(1/\log x)$. Also $\int_2^x dt/\log t = x/\log x + \int_2^x dt/\log^2 t + O(1)$ and $\int_2^x dt/(t \log t) = \log \log x + O(1)$. Absorbing all lower-limit constants into B gives $\sum_{p \leq x} f(p) = ax + (a+c)(x/\log x + \int_2^x dt/\log^2 t) + b \log \log x + B + O(1/\log x)$.

Exercise 2. Show that for $n \geq 16$, $\sum_{p|n} 1/p \ll \log \log \log n$. *Solution.* The sum is maximized, for a fixed size of n , by using the smallest possible primes. If $p_1, \dots, p_k$ are the distinct prime divisors of n and $y = p_k$, then $\prod_{p \leq y} p \leq n$ after decreasing the primes if necessary, so Chebyshev's lower bound for $\vartheta(y) = \sum_{p \leq y} \log p$ gives $y \ll \log n$. Therefore $\sum_{p|n} 1/p \leq \sum_{p \leq C \log n} 1/p = \log \log \log n + O(1)$ by Mertens' theorem.

Exercise 3. Show that $\sum_p 1/(p(\log \log p)^\alpha)$ converges for $\alpha > 1$ and diverges for $\alpha \leq 1$. *Solution.* Ignore the finitely many primes with $\log \log p \leq 0$. By Mertens' theorem, $A(x) = \sum_{p \leq x} 1/p = \log \log x + B + O(1/\log x)$. Abel summation reduces the series to the integral $\int_2^\infty u^{-\alpha} du$ with $u = \log \log t$. This integral converges exactly for $\alpha > 1$ and diverges for $\alpha \leq 1$.

Exercise 4. For $S(x) = \#\{n \leq x : p \mid n \Rightarrow p \leq \sqrt{n}\}$, prove $S(x) = (1 - \log 2)x + O(x/\log x)$. *Solution.* The excluded integers have a unique prime factor $p > \sqrt{n}$, so they are $n = mp$ with $m < p$ and $mp \leq x$. Hence their number is $\sum_{m \leq \sqrt{x}} \{\pi(x/m) - \pi(m)\}$. By the prime number theorem, uniformly for $m \leq \sqrt{x}$, $\pi(x/m) = x/(m \log(x/m)) + O(x/(m \log^2(x/m)))$, while $\sum_{m \leq \sqrt{x}} \pi(m) \ll x/\log x$. Therefore

the excluded number is $x \sum_{m \leq \sqrt{x}} 1/(m \log(x/m)) + O(x/\log x) = x \int_1^{\sqrt{x}} dt/(t \log(x/t)) + O(x/\log x) = x \log 2 + O(x/\log x)$, and the result follows.

Exercise 5. Show that, as $x \rightarrow \infty$, the assertion $\sum_{n \leq x} \mu(n)/n = o(1)$ would imply the prime number theorem. *Solution.* Let $A(x) = \sum_{n \leq x} \mu(n)/n$. Abel summation gives $M(x) = \sum_{n \leq x} \mu(n) = xA(x) - \int_1^x A(t) dt = o(x)$. The course equivalence between $M(x) = o(x)$ and the prime number theorem then yields $\psi(x) \sim x$, hence $\pi(x) \sim x/\log x$.

Exercise 6. Let $L(N) = [1, 2, \dots, N]$ be the least common multiple of $1, \dots, N$. Show that the prime number theorem is equivalent to $\lim_{N \rightarrow \infty} L(N)^{1/N} = e$. *Solution.* For each prime power $p^a \leq N$, the lcm contains one factor p , and these are all its prime factors counted with multiplicity. Thus $\log L(N) = \sum_{p^a \leq N} \log p = \psi(N)$. Therefore $L(N)^{1/N} \rightarrow e$ iff $\psi(N)/N \rightarrow 1$, which is one of the standard equivalent forms of the prime number theorem.

Exercise 7. For $n > 1$, show that some prime p satisfies $p \parallel n!$; recall that $p^r \parallel m$ means $p^r \mid m$ but $p^{r+1} \nmid m$. *Solution.* By Bertrand's postulate, there is a prime p with $n/2 < p \leq n$. Then p occurs among $1, \dots, n$, but $2p > n$, so $p^2 \nmid n!$. Hence $p \parallel n!$.

Exercise Sheet 4

Exercise 1. For an arithmetic function f with Dirichlet series $L_f(s)$, prove the equivalence: (a) $f(n) \ll n^A$ for some $A > 0$ and all $n \geq 1$; (b) the abscissa of absolute convergence of $L_f(s)$ is finite. *Solution.* If $|f(n)| \ll n^A$, then $\sum |f(n)|n^{-\sigma}$ converges for $\sigma > A + 1$, so the absolute abscissa is finite. Conversely, if the series is absolutely convergent at σ_0 , then the terms $|f(n)|n^{-\sigma_0}$ are bounded, hence $|f(n)| \ll n^{\sigma_0}$.

Exercise 2. Construct examples: (a) find f with $\sigma_c = \sigma_a - 1$; (b) for real σ_a, σ_c with $\sigma_a - 1 \leq \sigma_c \leq \sigma_a$, find f such that $L_f(s)$ has simple abscissa σ_c and absolute abscissa σ_a . *Solution.* (a) The eta series $\sum_{n \geq 1} (-1)^{n-1} n^{-s}$ has $\sigma_c = 0$ and $\sigma_a = 1$, so $\sigma_c = \sigma_a - 1$.

(b) Suppose $\sigma_a - 1 \leq \sigma_c \leq \sigma_a$. Define $f(n) = (-1)^{n-1} n^{\sigma_a-1} + \mathbf{1}_{n=2^k} (k \geq 0) n^{\sigma_c}$. The first series has simple abscissa $\sigma_a - 1$ and absolute abscissa σ_a ; the lacunary geometric series has both abscissae equal to σ_c . Therefore the sum has σ_c as its simple abscissa and σ_a as its absolute abscissa.

Exercise 3. Find the absolute abscissa of $F(s) = \sum_{m_1, m_2, m_3 \geq 1} (m_1 + m_2 + m_3)^{-s}$ and express it using $\zeta(s)$. *Solution.* For $n = m_1 + m_2 + m_3$, the number of positive triples is $\binom{n-1}{2}$. Hence $F(s) = \sum_{n \geq 3} \binom{n-1}{2} n^{-s} = \frac{1}{2}(\zeta(s-2) - 3\zeta(s-1) + 2\zeta(s))$. The coefficients are $\asymp n^2$, so the abscissa of absolute convergence is 3.

Exercise 4. Let $\omega(n), \Omega(n)$ be the numbers of distinct and total prime factors, and if $n = p_1^{a_1} \cdots p_k^{a_k}$ set $\kappa(n) = a_1 \cdots a_k$, $\kappa(1) = 1$. Express $\sum_{n \geq 1} \tau(n^2) n^{-s}$, $\sum_{n \geq 1} 2^{\omega(n)} n^{-s}$, $\sum_{n \geq 1} (-1)^{\Omega(n)} n^{-s}$, and $\sum_{n \geq 1} 3^{\omega(n)} \kappa(n) n^{-s}$ in terms of $\zeta(s)$. *Solution.* Euler factors give $\sum_{n \geq 1} \tau(n^2) n^{-s} = \prod_p (1 + p^{-s})(1 + p^{-s})^{-2} = \zeta(s)^3/\zeta(2s)$ and $\sum_{n \geq 1} 2^{\omega(n)} n^{-s} = \prod_p (1 + p^{-s})(1 + p^{-s})^{-1} = \zeta(s)^2/\zeta(2s)$. Also $\sum_{n \geq 1} (-1)^{\Omega(n)} n^{-s} = \prod_p (1 + p^{-s})^{-1} = \zeta(2s)/\zeta(s)$, and since $1 + 3 \sum_{a \geq 1} aX^a = (1 + X + X^2)/(1 - X)^2 = (1 - X^3)/(1 - X)^3$, $\sum_{n \geq 1} 3^{\omega(n)} \kappa(n) n^{-s} = \zeta(s)^3/\zeta(3s)$.

Exercise 5. For $c > 0$ write $\int_{(c)} = \int_{c-i\infty}^{c+i\infty}$. Show $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} ds = 0$ if $0 < y \leq 1$ and $= \log y$ if $y \geq 1$; then express $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} (-\zeta'/\zeta)(s) ds$ as a finite sum involving Λ . *Solution.* (a) Move the line of integration. If $0 < y < 1$, shift to the right and get 0; if $y > 1$, shift to the left and pick the double pole of y^s/s^2 at $s = 0$, whose residue is $\log y$. At $y = 1$ the value is 0.

(b) Since $-\zeta'(s)/\zeta(s) = \sum_{n \geq 1} \Lambda(n) n^{-s}$ for $\text{Re}(s) > 1$, termwise integration and (a) give $\frac{1}{2\pi i} \int_{(2)} y^s s^{-2} (-\zeta'/\zeta)(s) ds = \sum_{n \leq y} \Lambda(n) \log(y/n)$.

Exercise 6. Derive, for $\text{Re } s > 1$, $1/\zeta(s) = s \int_1^\infty M(x) x^{-s-1} dx$ with $M(x) = \sum_{n \leq x} \mu(n)$, and $-\zeta'(s)/\zeta(s) = s \int_1^\infty \psi(x) x^{-s-1} dx$ with $\psi(x) = \sum_{n \leq x} \Lambda(n)$. *Solution.* For $\text{Re}(s) > 1$, Abel summation in Stieltjes form gives $\sum a(n) n^{-s} = s \int_1^\infty A(x) x^{-s-1} dx$, where $A(x) = \sum_{n \leq x} a(n)$, provided the boundary terms vanish. Taking $a = \mu$ gives $1/\zeta(s) = s \int_1^\infty M(x) x^{-s-1} dx$, and taking $a = \Lambda$ gives $-\zeta'(s)/\zeta(s) = s \int_1^\infty \psi(x) x^{-s-1} dx$.

Exercise Sheet 5

Exercise 1. Let $k = 2^\alpha$ and write odd a as $a \equiv (-1)^{r-1(a)} 5^{r_0(a)} \pmod{2^\alpha}$. For $\alpha \geq 2$, prove (a) $r_{-1}(a) \equiv (a-1)/2 \pmod{2}$; (b) $(-1)^{r'-1} 5^{r'_0} \equiv a \pmod{2^\alpha}$ iff $r'_{-1} \equiv r_{-1}(a) \pmod{2}$ and $r'_0 \equiv r_0(a) \pmod{2^{\alpha-2}}$. *Solution.* (a) Since $5^{r_0(a)} \equiv 1 \pmod{4}$, the congruence $a \equiv (-1)^{r-1(a)} 5^{r_0(a)} \pmod{2^\alpha}$ gives $a \equiv (-1)^{r-1(a)} \pmod{4}$. For odd a , this is equivalent to $r_{-1}(a) \equiv (a-1)/2 \pmod{2}$.

(b) If the two index congruences hold, the residues are clearly equal. Conversely, if $(-1)^{r'-1} 5^{r'_0} \equiv a \equiv (-1)^{r-1} 5^{r_0} \pmod{2^\alpha}$, reduction modulo 4 gives $r'_{-1} \equiv r_{-1} \pmod{2}$. Then $5^{r'_0-r_0} \equiv 1 \pmod{2^\alpha}$, and since 5 has order $2^{\alpha-2}$ modulo 2^α , $r'_0 \equiv r_0 \pmod{2^{\alpha-2}}$.

Exercise 2. Let $\varphi^*(k)$ be the number of primitive characters modulo k . Prove $\varphi^*(k) = \sum_{d|k} \mu(d) \varphi(k/d) = k \prod_{p|k} (1-2/p) \prod_{p^2|k} (1-1/p)^2$. *Solution.* Every character modulo k is induced from a unique primitive character whose conductor divides k , so $\varphi(k) = \sum_{d|k} \varphi^*(d)$. Möbius inversion gives $\varphi^*(k) = \sum_{d|k} \mu(d) \varphi(k/d)$. The right hand side is multiplicative in k . For $k = p$, it is $\varphi(p) - \varphi(1) = p - 2 = p(1 - 2/p)$; for $k = p^a$, $a \geq 2$, it is $\varphi(p^a) - \varphi(p^{a-1}) = p^{a-2}(p-1)^2 = p^a(1 - 1/p)^2$. Multiplying the local factors gives $k \prod_{p|k} (1 - 2/p) \prod_{p^2|k} (1 - 1/p)^2$.

Exercise 3. Let $\chi \pmod{k}$ and $k^* \mid k$. Prove that χ is induced by a character modulo k^* iff $\chi(n_1) = \chi(n_2)$ whenever $(n_i, k) = 1$ and $n_1 \equiv n_2 \pmod{k^*}$; equivalently, iff $\chi(n) = 1$ whenever $(n, k) = 1$ and $n \equiv 1 \pmod{k^*}$. *Solution.* If χ is induced from $\chi^* \pmod{k^*}$, then $\chi(n_1) = \chi(n_2)$ whenever $(n_i, k) = 1$ and $n_1 \equiv n_2 \pmod{k^*}$. Conversely, this condition lets us define $\chi^*(a) = \chi(n)$ for any lift $n \equiv a \pmod{k^*}$ with $(n, k) = 1$; such a lift exists by CRT, and the condition makes the definition independent of the lift. Thus χ is induced by χ^* . For units modulo k , the first condition is equivalent to $\chi(n_1 n_2^{-1}) = 1$ whenever $n_1 \equiv n_2 \pmod{k^*}$. Taking $n_2 = 1$ gives the second criterion, and the second criterion implies the first by applying it to $n_1 n_2^{-1} \equiv 1 \pmod{k^*}$.

Exercise 4. For $k = 2^\alpha$, $\alpha \geq 3$, define $\chi(n; k, m_{-1}, m_0) = 0$ if $(n, 2) > 1$, and $\chi(n; k, m_{-1}, m_0) = (-1)^{m_{-1}r-1(n)} e^{2\pi i m_0 r_0(n)/2^{\alpha-2}}$ if $(n, 2) = 1$. Prove that this character is primitive iff $(m_0, 2) = 1$. *Solution.* If m_0 is even, then $e^{2\pi i m_0 r_0/2^{\alpha-2}} = e^{2\pi i (m_0/2) r_0/2^{\alpha-3}}$, so the value of χ only depends on $r_{-1} \pmod{2}$ and $r_0 \pmod{2^{\alpha-3}}$, i.e. only on $n \pmod{2^{\alpha-1}}$. Hence χ is induced from modulus $2^{\alpha-1}$ and is not primitive. If m_0 is odd, let $n \equiv 5^{2^{\alpha-3}} \equiv 1 + 2^{\alpha-1} \pmod{2^\alpha}$. Then $n \equiv 1 \pmod{2^{\alpha-1}}$, but $\chi(n) = e^{2\pi i m_0/2} = -1$. Thus χ is not induced from $2^{\alpha-1}$, and therefore not from any smaller divisor of 2^α . It is primitive.

Exercise 5. Verify that the only real primitive character modulo 4 is $\chi_4(n) = 0$ if $(n, 2) > 1$ and $(-1)^{(n-1)/2}$ otherwise; and the real primitive characters modulo 8 are $\chi_8(n) = 0$ if $(n, 2) > 1$ and $(-1)^{(n^2-1)/8}$ otherwise, and $\chi_4(n)\chi_8(n) = 0$ if $(n, 2) > 1$ and $(-1)^{(n-1)(n+5)/8}$ otherwise.

Solution. Modulo 4, the unit group has order 2, so the only non-principal real character is $\chi_4(n) = 0$ for even n and $\chi_4(n) = (-1)^{(n-1)/2}$ for odd n ; it is primitive. Modulo 8, the group of units is $\{\pm 1, \pm 5\} \simeq C_2 \times C_2$. By Exercise 4, primitive characters have m_0 odd, so the two real primitive ones are obtained from $(m_{-1}, m_0) = (0, 1)$ and $(1, 1)$. Evaluating on 1, 3, 5, 7 gives respectively $\chi_8(n) = (-1)^{(n^2-1)/8}$ and $\chi_4(n)\chi_8(n) = (-1)^{(n-1)(n+5)/8}$, with value 0 on even n .

Exercise 6. For $f: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$ define $\hat{f}(\alpha) = q^{-1/2} \sum_{a \pmod{q}} f(a) e^{2\pi i \alpha a/q}$. Prove the convolution/Parseval identities, compute $\hat{1}_I$ for $I = [A, B] \subset [0, q]$, show $\sum_{\alpha \pmod{q}} |\hat{1}_I(\alpha)| \ll \sqrt{q} \log q$, and deduce Polya-Vinogradov: for primitive $\chi \pmod{q}$, $\sum_{n=1}^N \chi(n) \ll \sqrt{q} \log q$.

Solution. With the stated sign convention, orthogonality gives the corrected identity $\sum_{a \pmod{q}} f(a) g(a) = \sum_{\alpha \pmod{q}} \hat{f}(\alpha) \hat{g}(-\alpha)$. Taking $g = \bar{f}$ gives Parseval: $\sum_a |f(a)|^2 = \sum_\alpha |\hat{f}(\alpha)|^2$.

For $I = [A, B]$ and $L = B - A + 1$, $\hat{1}_I(\alpha) = q^{-1/2} \sum_{n=A}^B e^{2\pi i \alpha n/q}$, hence $\hat{1}_I(0) = L/\sqrt{q}$ and, for $\alpha \neq 0$, $\hat{1}_I(\alpha) = q^{-1/2} e^{2\pi i \alpha A/q} (1 - e^{2\pi i \alpha L/q}) / (1 - e^{2\pi i \alpha/q})$.

$e^{2\pi i\alpha/q}$). Thus $|\widehat{1}_I(\alpha)| \leq q^{-1/2} \min(L, (2\|\alpha/q\|)^{-1})$, and summing over α gives $\sum_{\alpha \bmod q} |\widehat{1}_I(\alpha)| \ll \sqrt{q} \log q$.

If χ is primitive, its Fourier transform is a normalized Gauss sum: $\widehat{\chi}(\alpha) = q^{-1/2} \tau(\chi) \overline{\chi}(\alpha)$ for $(\alpha, q) = 1$ and 0 otherwise, so $|\widehat{\chi}(\alpha)| \leq 1$. Hence for every interval $I \bmod q$, $|\sum_{n \in I} \chi(n)| = |\sum_{\alpha} \widehat{1}_I(\alpha) \widehat{\chi}(-\alpha)| \ll \sqrt{q} \log q$. Complete periods have sum 0, so this applies to $\sum_{n \leq N} \chi(n)$ for all N .

Exercise 7. For prime p , define $(n/p) = 0$ if $n \equiv 0 \pmod{p}$, 1 if $n \equiv x^2 \pmod{p}$ for some x , and -1 otherwise. Show (n/p) is a multiplicative character on $(\mathbb{Z}/p\mathbb{Z})^\times$, and use Polya–Vinogradov to show the least quadratic non-residue modulo p is $O(\sqrt{p} \log p)$. *Solution.* For odd prime p , Euler’s criterion gives $(n/p) \equiv n^{(p-1)/2} \pmod{p}$ for $p \nmid n$, and the value is ± 1 ; hence $(mn/p) = (m/p)(n/p)$ on $(\mathbb{Z}/p\mathbb{Z})^\times$. If every $1 \leq n \leq N$ were a quadratic residue, then $\sum_{n \leq N} (n/p) = N$. By Polya–Vinogradov this is $O(\sqrt{p} \log p)$, so taking N a sufficiently large constant multiple of $\sqrt{p} \log p$ forces a non-residue below N .

Exercise Sheet 6

Exercise 1. Let G be a finite abelian group and let $\widehat{G} = \{\chi : G \rightarrow \mathbb{C}^\times : \chi \text{ is a character}\}$ be its dual group. Prove $G \simeq \widehat{\widehat{G}}$. *Solution.* By the structure theorem, $G \simeq \prod_j \mathbb{Z}/q_j\mathbb{Z}$. The dual of $\mathbb{Z}/q\mathbb{Z}$ is generated by $x \mapsto e^{2\pi i x/q}$ and is cyclic of order q , hence isomorphic to $\mathbb{Z}/q\mathbb{Z}$. Since characters of a product are products of characters, $\widehat{G} \simeq \prod_j \widehat{\mathbb{Z}/q_j\mathbb{Z}} \simeq G$.

Exercise 2. For $\operatorname{Re} s > 1$, $\zeta(s) = \sum_{n=1}^\infty n^{-s}$. (a) Use Euler–Maclaurin or Abel summation to show $\zeta(s) = 1/(s-1) + 1 - s \int_1^\infty \{\xi\} \xi^{-s-1} d\xi$, giving meromorphic continuation to $\operatorname{Re} s > 0$. (b) Show $\zeta(s) = 1/(s-1) + \gamma + O(|s-1|)$ near $s=1$, where $\gamma = \lim_{n \rightarrow \infty} (\sum_{k=1}^n 1/k - \log n)$.

Solution. The standard Abel-summation formula is $\zeta(s) = s \int_1^\infty [x] x^{-s-1} dx = \frac{1}{s-1} + 1 - s \int_1^\infty \{x\} x^{-s-1} dx$ for $\operatorname{Re}(s) > 1$. The last integral converges uniformly on compact subsets of $\operatorname{Re}(s) > 0$, so this gives a meromorphic continuation there with only a simple pole at $s=1$. Moreover, $\lim_{s \rightarrow 1} (\zeta(s) - 1/(s-1)) = \lim_{N \rightarrow \infty} (H_N - \log N) = \gamma$. Thus $\zeta(s) = 1/(s-1) + \gamma + O(|s-1|)$ near $s=1$.

Exercise 3. If $\prod_{p \leq x} (1-1/p)^{-1} = C \log x + O(1)$, show $C = e^\gamma$. *Solution.* Put $P(x) = \prod_{p \leq x} (1-1/p)^{-1}$ and $L = \log x$. Comparing Euler products at $s = 1 + 1/L$ by Abel summation gives the standard Mertens comparison $\zeta(1 + 1/L) = e^{-\gamma} P(x)(1 + o(1))$. Since $\zeta(1 + 1/L) \sim L$ and $P(x) \sim CL$, we get $1 = e^{-\gamma} C$, hence $C = e^\gamma$.

Exercise 4. Let $\chi \bmod q$ be non-principal and $x \geq 2$. Prove: (a) $\sum_{n \leq x} \chi(n) \log n/n = -L'(1, \chi) + O_\chi(\log x/x)$; (b) $\sum_{n \leq x} \chi(n) \log n/n = L(1, \chi) \sum_{m \leq x} \chi(m) \Lambda(m)/m + O_\chi(x^{-1} \sum_{m \leq x} \Lambda(m))$, hence $\sum_{m \leq x} \chi(m) \Lambda(m)/m = O_\chi(1)$; (c) $\sum_{p \leq x} \chi(p) \log p/p = O_\chi(1)$; (d) $\sum_{p \leq x} \chi(p)/p = b(\chi) + O_\chi(1/\log x)$, where $b(\chi) = \log L(1, \chi) - \sum_p \sum_{k \geq 2} \chi(p^k)/(kp^k)$, so $\sum_p \chi(p)/p$ converges. *Solution.* (a) Since the partial sums of χ are bounded, the tail of $-L'(1, \chi) = \sum_{n \geq 1} \chi(n) \log n/n$ is $O_\chi(\log x/x)$ by Abel summation. This gives the formula.

(b) The coefficient identity $\chi(n) \log n = \sum_{m|n} \chi(m) \Lambda(m)$ lets us change order. Since $\sum_{\ell \leq x/m} \chi(\ell)/\ell = L(1, \chi) + O_\chi(m/x)$, the left side in (a) equals $L(1, \chi) B(x) + O_\chi(1)$, where $B(x) = \sum_{m \leq x} \chi(m) \Lambda(m)/m$ and we used $\sum_{m \leq x} \Lambda(m) \ll x$. As $L(1, \chi) \neq 0$, we get $B(x) = O_\chi(1)$.

(c) The prime-power part with exponent ≥ 2 is absolutely bounded, so $\sum_{p \leq x} \chi(p) \log p/p = O_\chi(1)$.

(d) Abel summation applied to $A(t) = \sum_{p \leq t} \chi(p) \log p/p = O_\chi(1)$ gives convergence of $\sum_p \chi(p)/p$ and the tail $O_\chi(1/\log x)$. From the Euler product, $\log L(1, \chi) = \sum_p \chi(p)/p + \sum_p \sum_{k \geq 2} \chi(p^k)/(kp^k)$, so the limiting constant is $b(\chi) = \log L(1, \chi) - \sum_p \sum_{k \geq 2} \chi(p^k)/(kp^k)$.

Exercise 5. For $|\arg s| \leq \pi - \delta$, prove $\Gamma'(s)/\Gamma(s) = \log s - 1/(2s) + O_\delta(|s|^{-2})$. *Solution.* Stirling’s formula in the sector says $\log \Gamma(s) = (s - \frac{1}{2}) \log s - s + \frac{1}{2} \log(2\pi) + 1/(12s) + O_\delta(|s|^{-3})$. Apply Cauchy’s integral formula to differentiate this asymptotic expansion uniformly on a small circle around s contained in the same sector. The derivative is $\Gamma'(s)/\Gamma(s) = \log s - 1/(2s) + O_\delta(|s|^{-2})$.

Exercise Sheet 7

Exercise 1. For any $\alpha > 0$, construct an entire function f_α of order exactly α . Hint: use $f_\alpha(z) = \sum_{j \geq 1} z^j/\Gamma(1 + j/\alpha)$. *Solution.* Let $f_\alpha(z) = \sum_{j \geq 0} z^j/\Gamma(1 + j/\alpha)$. The coefficient formula for the order of an entire function gives $\rho = \limsup_{j \rightarrow \infty} j \log j / \log \Gamma(1 + j/\alpha)$. By Stirling, $\log \Gamma(1 + j/\alpha) = (j/\alpha) \log j + O(j)$, so $\rho = \alpha$.

Exercise 2. Are there entire functions of order 0 which are not polynomials? *Solution.* Yes. For example $f(z) = \sum_{j \geq 0} e^{-j^2} z^j$ is not a polynomial and is entire. Its order is $\limsup_{j \rightarrow \infty} j \log j / j^2 = 0$ by the same coefficient formula.

Exercise 3. Let $f(s)$ be an entire function of order α . For every $\varepsilon > 0$, show $\sum_\rho |\rho|^{-\alpha-\varepsilon}$ converges, where ρ runs over the zeros of f , counted with multiplicity. *Solution.* Zeros at 0 are finite and may be omitted. Since f has order α , for every $\varepsilon > 0$ one has $\log M(r) \ll r^{\alpha+\varepsilon/2}$ for large r . Jensen’s formula gives $N(r) = \#\{\rho : |\rho| \leq r\} \ll r^{\alpha+\varepsilon/2}$. Therefore the dyadic estimate $\sum_{2^j \leq |\rho| < 2^{j+1}} |\rho|^{-\alpha-\varepsilon} \leq N(2^{j+1}) 2^{-j(\alpha+\varepsilon)} \ll 2^{-j\varepsilon/2}$ proves convergence.

Exercise 4. Using the logarithmic derivative of $\xi(s)$ and $\xi(s) = \xi(1-s)$, show that the constant B in $\xi(s) = e^{Bs} \prod_\rho (1-s/\rho) e^{s/\rho}$ satisfies $B = -\sum_\rho 1/\rho = -2 \sum_{\gamma > 0} \beta/(\beta^2 + \gamma^2)$, where $\rho = \beta + i\gamma$ and $\sum_\rho 1/\rho := \lim_{x \rightarrow \infty} \sum_{|\rho| \leq x} 1/\rho$. *Solution.* The genus-one product gives $(\zeta^*)'(s)/\zeta^*(s) = B + \sum_\rho (1/(s-\rho) + 1/\rho)$. The functional equation pairs the zeros symmetrically as ρ and $1-\rho$; comparing this genus-one product with the symmetric genus-zero product gives $B = -\lim_{R \rightarrow \infty} \sum_{|\rho| \leq R} 1/\rho$. Pairing conjugate zeros $\rho = \beta + i\gamma$ and $\bar{\rho} = \beta - i\gamma$ then gives $\sum_\rho 1/\rho = 2 \sum_{\gamma > 0} \beta/(\beta^2 + \gamma^2)$, hence $B = -\sum_\rho 1/\rho = -2 \sum_{\gamma > 0} \beta/(\beta^2 + \gamma^2)$.

Exercise 5. Let χ be primitive modulo q with $\chi(-1) = 1$, and put $\Lambda_\chi(s) = \pi^{-s/2} q^{s/2} \Gamma(s/2) L(s, \chi)$. Show $L(s, \chi) = \sum_{n \geq 1} \chi(n) n^{-s}$ continues entire and $\Lambda_\chi(1-s) = (\sqrt{q}/\tau(\chi)) \Lambda_\chi(s)$. What goes wrong when $\chi(-1) = -1$? Hint: use $\theta(x, \chi) = \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 x/q}$, $2\Lambda_\chi(s) = \int_0^\infty \theta(x, \chi) x^{s/2} dx/x$, and $\theta(1/x, \chi) = \sqrt{xq} \theta(x, \chi)/\tau(\chi)$. *Solution.* For $\operatorname{Re}(s) > 1$, evenness gives $2\Lambda_\chi(s) = \int_0^\infty \theta(x, \chi) x^{s/2} dx/x$. Split the integral at 1 and in $(0, 1)$ use $x \mapsto 1/x$ together with $\theta(1/x, \chi) = \sqrt{xq} \theta(x, \chi)/\tau(\chi)$. The result is a sum of two integrals over $[1, \infty)$ with exponential decay, hence an entire continuation, and the comparison of the two halves gives $\Lambda_\chi(1-s) = (\sqrt{q}/\tau(\chi)) \Lambda_\chi(s)$. If $\chi(-1) = -1$, then the even theta series $\theta(x, \chi)$ is identically 0 because the terms n and $-n$ cancel. One must instead use the odd theta series with an extra factor n , and the gamma factor becomes $\Gamma((s+1)/2)$ rather than $\Gamma(s/2)$.

Exercise Sheet 8

Exercise 1. Assume there is $C > 0$ such that every zero ρ of $\zeta(s)$ satisfies $\operatorname{Re} \rho < 1 - C/\log(2 + |\operatorname{Im} \rho|)$. Without the explicit formula, show there is $c > 0$ such that $\psi(x) = \sum_{n \leq x} \Lambda(n) = x + O(xe^{-c\sqrt{\log x}})$. Hint: use Perron’s formula and shift from $\sigma = 1 + 1/\log x$ to $\sigma = 1 - C/(2 \log T)$ with optimal T . *Solution.* It is enough to treat non-integral x , since changing x by $O(1)$ changes $\psi(x)$ by $O(\log x)$. Put $L = \log x$, $b = 1 + 1/L$, and $a = 1 - C/(2 \log T)$, where $3 \leq T \leq x$ will be chosen. Truncated Perron gives $\psi(x) = (2\pi i)^{-1} \int_{b-iT}^{b+iT} -(\zeta'/\zeta)(s) x^s s^{-1} ds + O(x(\log x)^2/T + \log x)$. The rectangle $a \leq \operatorname{Re} s \leq b$, $|\operatorname{Im} s| \leq T$, contains no zeros of ζ and only the pole at $s=1$. On this rectangle the usual logarithmic-derivative estimate gives $-\zeta'/\zeta(s) \ll (\log T)^2$: indeed Lemma 10.1 reduces the estimate to zeros with $|\operatorname{Im} \rho - \operatorname{Im} s| < 1$, there are $O(\log T)$ of them, and the zero-free region gives distance $\gg 1/\log T$. By Cauchy’s theorem, shifting the contour crosses the residue x at $s=1$, so $\psi(x) - x \ll x^a (\log T)^3 + x(\log T)^2/T + x(\log x)^2/T + \log x$. Taking $T = \exp(\sqrt{L})$ gives $x^a = x \exp(-C\sqrt{L}/2)$ and $x/T = xe^{-\sqrt{L}}$; the powers of $\log x$ are absorbed by decreasing the constant in the exponential. Hence $\psi(x) = x + O(xe^{-c\sqrt{\log x}})$ for some $c > 0$.

Exercise 2. Let $\mu(n)$ be the Mobius function. Apply Perron’s formula to show $\sum_{n \leq x} \mu(n) = O(xe^{-c\sqrt{\log x}})$ for some $c > 0$. You may use: if

$\zeta(\sigma + i\tau) \neq 0$ for $\sigma \geq 1 - c/\log(|\tau| + 2)$, then $1/|\zeta(\sigma + i\tau)| \ll \log(|\tau| + 2)$ when $\sigma \geq 1 - c/(2\log(|\tau| + 2))$. *Solution.* Let $M(x) = \sum_{n \leq x} \mu(n)$, $L = \log x$, $b = 1 + 1/L$, and $a = 1 - c_0/(2\log T)$, where c_0 is the zero-free-region constant appearing in the stated fact. Perron's formula for $1/\zeta(s) = \sum \mu(n)n^{-s}$ gives $M(x) = (2\pi i)^{-1} \int_{b-iT}^{b+iT} x^s/(s\zeta(s)) ds + O(x \log x/T + 1)$. In the rectangle $a \leq \operatorname{Re} s \leq b$, $|\operatorname{Im} s| \leq T$, the function $1/\zeta(s)$ is holomorphic; at $s = 1$ it even has a zero, not a pole. The given fact gives $1/|\zeta(s)| \ll \log T$ on the new vertical side. Therefore the contour shift gives $M(x) \ll x^a (\log T)^2 + x \log T/T + x \log x/T + 1$. With $T = \exp(\sqrt{\log x})$, this is $O(xe^{-c\sqrt{\log x}})$ after decreasing $c > 0$.

Exercise 3. For any $N \in \mathbb{N}$, show $\operatorname{Li}(x) = \frac{x}{\log x} \sum_{n=0}^{N-1} \frac{n!}{(\log x)^n} + O_N(x/(\log x)^{N+1})$. In particular, this explains why one cannot have $\pi(x) = x/\log x + O(x/(\log x)^{2.1})$. *Solution.* Set $I_k(x) = \int_2^x (\log u)^{-k-1} du$. Since $\frac{d}{du}(u/(\log u)^{k+1}) = (\log u)^{-k-1} - (k+1)(\log u)^{-k-2}$, we have $I_k(x) = x/(\log x)^{k+1} + (k+1)I_{k+1}(x) + O_k(1)$. Iterating this identity from $k = 0$ to $N-1$ gives $\operatorname{Li}(x) = \sum_{n=0}^{N-1} n!x/(\log x)^{n+1} + N!I_N(x) + O_N(1)$. Finally $I_N(x) \ll_N x/(\log x)^{N+1}$, which proves the formula.

Exercise 4. Show that $\sum_{n \leq x} \mu(n) = O(x^{1-\theta+\varepsilon})$ if and only if $\sum_{n \leq x} \Lambda(n) = x + O(x^{1-\theta+\varepsilon})$. *Solution.* Write $\alpha = 1 - \theta$, and interpret the estimates as holding for every $\varepsilon > 0$. For $\operatorname{Re} s > 1$, $1/\zeta(s) = s \int_1^\infty M(u)u^{-s-1} du$ and $-\zeta'(s)/\zeta(s) = s/(s-1) + s \int_1^\infty R(u)u^{-s-1} du$, where $M(u) = \sum_{n \leq u} \mu(n)$ and $R(u) = \psi(u) - u$.

If $M(u) = O(u^{\alpha+\varepsilon})$ for every $\varepsilon > 0$, the first integral continues $1/\zeta(s)$ holomorphically to $\operatorname{Re} s > \alpha$. Hence $\zeta(s)$ has no zeros there. Applying Perron's formula to $-\zeta'/\zeta(s)$ and shifting the line to $\operatorname{Re} s = \alpha + \varepsilon/2$, the only crossed pole is the pole at $s = 1$, with residue x ; the standard zero-free-line bounds for ζ'/ζ give $\psi(x) = x + O(x^{\alpha+\varepsilon})$.

Conversely, if $R(u) = O(u^{\alpha+\varepsilon})$, the second integral shows that $-\zeta'/\zeta(s)$ is meromorphic in $\operatorname{Re} s > \alpha$ with only the pole at $s = 1$. Thus $\zeta(s)$ has no zero in this half-plane. Perron's formula for $1/\zeta(s)$, shifted to $\operatorname{Re} s = \alpha + \varepsilon/2$, then gives $M(x) = O(x^{\alpha+\varepsilon})$. Replacing α by $1 - \theta$ proves the equivalence.